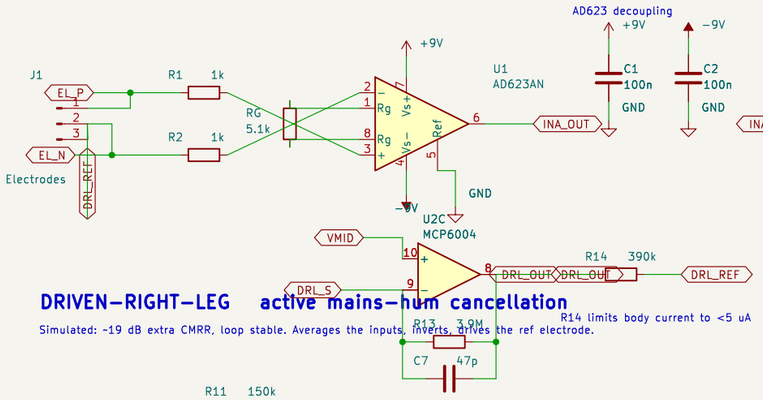
STAGE 1 INSTRUMENTATION AMPLIFIER G1 = 1 + 100k/RG = 21

Inputs via 1k protection (1% for CMRR). Low gain: electrode offset can reach +/−300 mV.

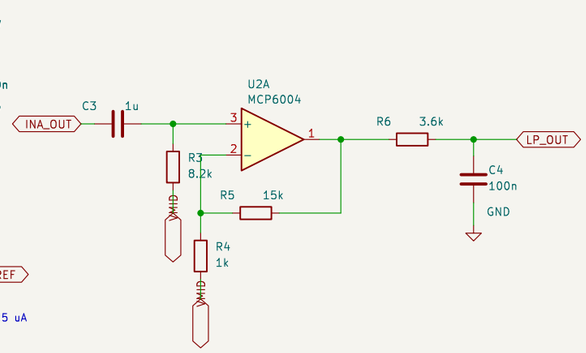


DRIVEN−RIGHT−LEG active mains−hum cancellation

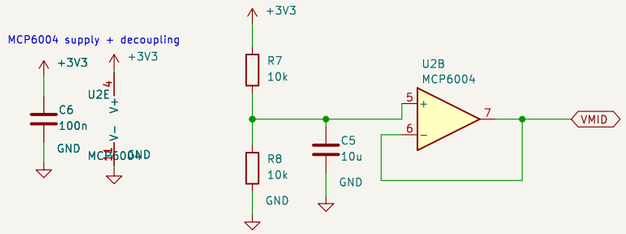
Simulated: ~19 dB extra CMRR, loop stable. Averages the inputs, inverts, drives the ref electrode.



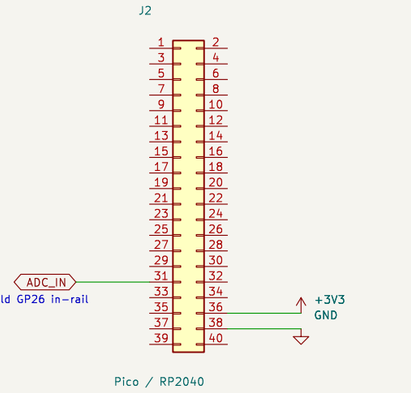
STAGE 2/3 HP 20Hz −> GAIN 16 −> LP 442Hz (R5=15k default; raise for weak signals)



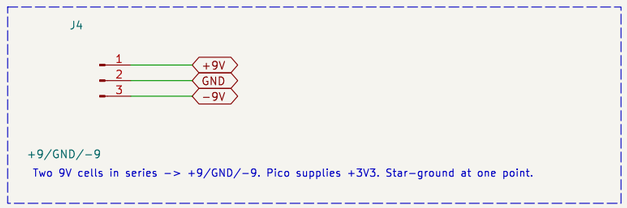
VMID = 1.65 V reference (buffered)



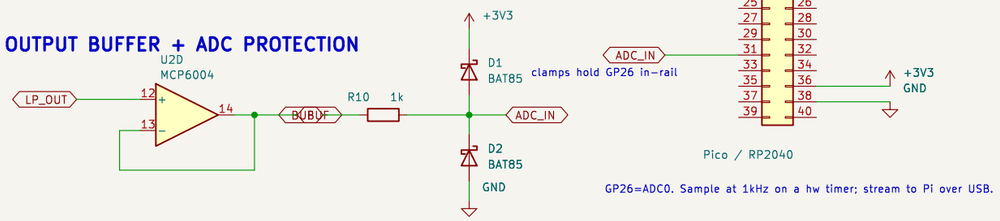
ACQUISITION



SUPPLY − battery only (off−board)



OUTPUT BUFFER + ADC PROTECTION



Gain 21 x 48 = 1000. Pico samples ADC0 at 1 kHz on a hardware timer.
BATTERY POWER ONLY − never wear electrodes with mains equipment connected

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Title: EMG Front End + Acquisition

Size: A3	Date:	Rev: C
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